Problem Set 4

CSCI 3271 – Operating Systems
2026.10.30 – Tim Price

1. A FAT32 file system is configured with a disk-block size of 2 KiB on a 650-MiB disk partition. The directory `C:\Documents\CSCI3271\` contains 2 files: Syllabus.doc of size 19 KiB and Textbook.pdf of size 5 MiB.

   1.A. How large is the FAT for this file system, in bytes?

   There would be `650 * 1024/2 = 332800` total clusters. Fat32 would use 4 bytes-per-entry so `332800 * 4 = 1331200` bytes.

   1.B. How many disk blocks are used by the directory and the two files? How long is each file's linked list in the FAT?

   - directory: 1 block ~ estimate
   - syllabus.doc: `19/2` (rounded up) `= 10` blocks
   - textbook.pdf: `5 Mib = 5120 Kib` then `5120 / 2 = 2560` blocks
   - total: 2571 blocks
   - linked list: each files linked-list is as long as its block count. So syllabus has a length of 10 and textbook has a length of 2560.

   1.C. Suppose a user process writes 5 KiB of data to the end of Syllabus.doc. Describe how the file system would change as a result of this operation. Specify all the data structures and blocks that would change, and how many blocks the file uses afterwards.

   - Initially the syllabus file has 10 blocks, now it should grow to `24 / 2 = 12` blocks.
   - The syllabus block already had leftover room at its end because of rounding, so it would fill that space before adding an additioal 2 blocks.
   - The files linked list is physically seperated but the pointer in the last block should now be pointing to the head block in the newly allocated 2 block section.
   - The FAT entry will be updated to identify the end of the linked-list as the end of the second logical block.
   - The directory entry should be updated accordingly (file size updated).

2. A file system has a disk-block size of 2 KiB, and block addresses require 4 bytes. The blocks that make up a file are pointed to by an i-node which contains 4 direct addresses, one single-indirect block entry, and one double-indirect block entry. What is the largest possible file size (in KiB) in this system?

   - Each block adress requires 4 bytes so 2KiB is 2048 bytes `2048 / 4 = 512` addresses for each block.
   - The i-node can adress using 4 direct pointers, 512 single-indirect pointers, or `512 * 512 = 262144` double-indirect pointers.
   - total: 262660 blocks which is `262660 * 2 KiB = 525320 KiB`.

3. A Linux ext2 file system is configured with a disk-block size of 4KiB and uses block addresses of 4 bytes. The directory `/home/etremel/` contains 2 files: research-notes.txt of size 9.4 KiB, and desktop-wallpaper.png of size 10.5 MiB.

   3.A. Describe the set of blocks needed to store the directory and files on the disk. How many blocks in total do they occupy?

   - Directory: needs an i-node and at least 1 data block.
   - research-notes.txt: `9.4 KiB / 4 KiB = 3` (rounded up) data blocks.
   - desktop-wallpaper.png: if `10.5 MiB * 1024 = 10752 KiB` then `10752 KiB / 4 KiB = 2688` data blocks (rounded up).
   - total: `1 + 3 + 2688 = 2696` total data blocks.

   3.B. Suppose a user process writes 4.5KiB of data to the end of research-notes.txt. Describe how the file system would change as a result of this operation, compared to its state in part A. Specify all the

data structures and blocks that would change, and how many blocks the file uses afterwards.

- `9.4 KiB + 4.5 KiB = 13.9 KiB` which now will need `13.9 / 4 = 4` (rounded up) blocks.
- The original blocks had some free space which is now full, and an additional block of space added.
- The i-node will use another direct pointer,
- The file size will be updated but the directory entry probably stays the same as the i-node number remained the same.

4. A computer has two memory-mapped I/O devices that it can use to display output:

   A text-based terminal and a graphical display adapter.

   The text-based terminal is controlled using a character buffer, which has one entry per character that can be displayed on the screen. The terminal can display 25 lines of 80 characters each.

   The graphical display adapter has a video display buffer that contains the image currently displayed on the screen. The screen has a resolution of 1280 × 1024 and uses 16-bit color.

   4.A Assuming it takes the computer 4 nanoseconds to copy a byte of memory, how long does it take to completely rewrite each type of display?

   - text-based terminal: `25 * 80 = 2000` total bytes. `2000 * 4 ns = 8000 ns` of total time to refresh every byte ~ 8 microseconds
   - graphical display adapter: `1280 * 1024 = 1310720 pixels` which makes `1310720 * 2 bytes = 2621440 bytes` total. Then, `2621440 * 4 = 10485760` ns ~ 10.48.. milliseconds

   4.B. Imagine both of these display devices could be used to show the user a command shell, which needs to display the output from each command a user types by scrolling all the texton the screen up one line at a time. Which device would you prefer to use, and why?

   Well it takes `10485760 / 8000` seconds longer to rewrite the graphical display adapters screen. A shell even if just a single line was written on a full display would require the entire screen to be rewritten. So, it would be much faster to use the text-based terminal, especially since graphics are hardly a concern when working in a shell.